

WEEK FIVE

Modelling Requirements

This week's lecture aims cover the following:

- *Introduction to Modelling*
- *User Stories*
- *UML: Categories and Diagrams*
- *UML Use case diagram*
- *UML Sequence diagram*
- *UML State diagram*
- *Class Examples/Discussions*

Modelling Requirements

Modelling

Modelling is an essential task in any software project, especially big projects

- A model is an abstraction of an aspect of an existing or proposed system
- They can be used to present a layman description of different aspects of the system
- The models are created before the Implementation (coding) phase starting from Requirements phase till Detailed Design
- They are used for verifying that all the requirements are present and correctly modelled.
- Also used in the Verification and Validation phase

Modelling Requirements

Use of Modelling

Models for user: (logical model) for user capabilities, characteristics, and product perspectives

- Showing:
 - The requirements that the system should implement
 - An external view on the system

Models for design: (physical model)

- Showing:
 - The architecture of the system
 - The distribution of system processes and their synchronization
 - The physical components and the redistribution of the software components and the (physical) hardware components.

User Stories

What are User Stories?

- A User Story refers to a couple of sentences written to express a specific user need
- They attempt to talk about the requirement in a narrative, rather than just write the requirements.
- They describe features from the view of the user or customer who desires the capability
- User stories MAY follow the syntax:
 - As a *<user category>*, I want to *<action description>*, so that *<reason>*

E.g. As a Student, I can unblock my registration, so as to update my course selection for resubmission

User Stories

What are User Stories?

- They are often written on index cards or sticky notes, or excel cells
- They come in handy for agile-based methodologies, because it enables developers to work with system users through out the development project
- It contains enough information for the user to understand the functionality to be accomplished
- It does not necessarily need to be a concise description of how the system works

User Stories

User Story Examples

Course Selection

A student wants to generate his or her course form for the semester, so he or she checks the boxes for his or her semester courses, from the displayed list of courses

Students can only get discounts on seminar fees for which they have prerequisites

168 Search by Name

As a help desk operator I want to search for my customers by their first and last names so that customer response times remain short

As a	I want	So that
Customer	To see total cost of items in my shopping cart, while shopping	I can work with my budget
Seller	To add items to my page	Customers can see the items available for sale

What is UML?

- UML - Unified Modelling Language
- UML is a standardized general-purpose modelling language in the field of software engineering. The standard is managed, and was created by the Object Management Group. OMG.
- UML includes a set of graphical notation techniques to create visual models of software-intensive systems. It is used for software specification and documenting

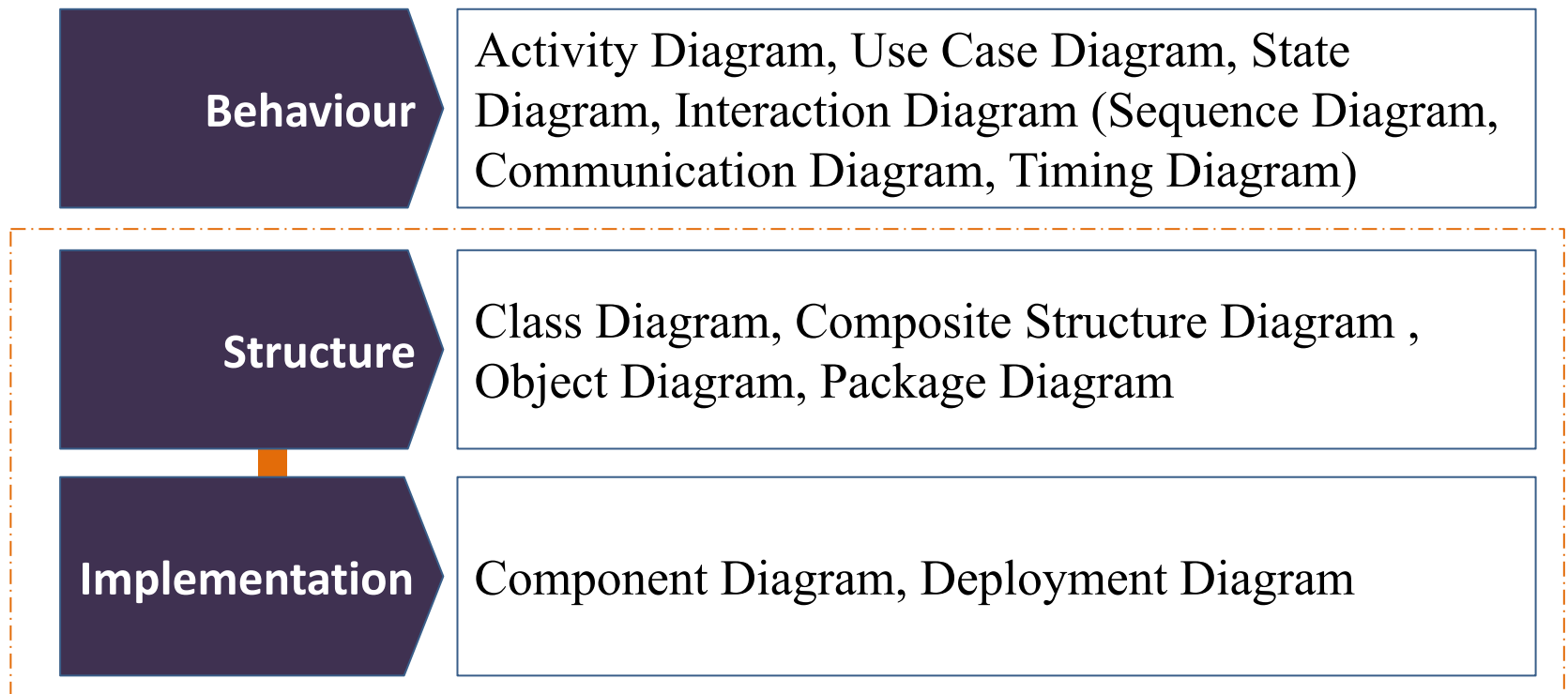
UML Versions:

1.0	(01/1997)
1.3	(03/2000)
1.4	(09/2001)
1.5	(03/2003)
2.0	(07/2005)
2.1.1	(08/2007)
2.1.2	(11/2007)
2.2	(02/2009)
2.3	(05/2010)
2.4	(03/2011)
2.4.1	(08/2011)
2.5	(06/2015)

- UML provides a standard way to depict the analysis carried out on the intended software solution and to visualize the design of the system
- It is used by business analysts, software architects and technical developers for describing, specifying, designing, and documenting existing or new business processes, structure and behaviour of software systems
- It can also used by the validation and verification team

Categorizing UML Diagrams

UML Diagrams are in different forms, belonging to sub-categories of the entire UML scope, and modelling specific aspects of a software system.



UML

Categorizing UML Diagrams

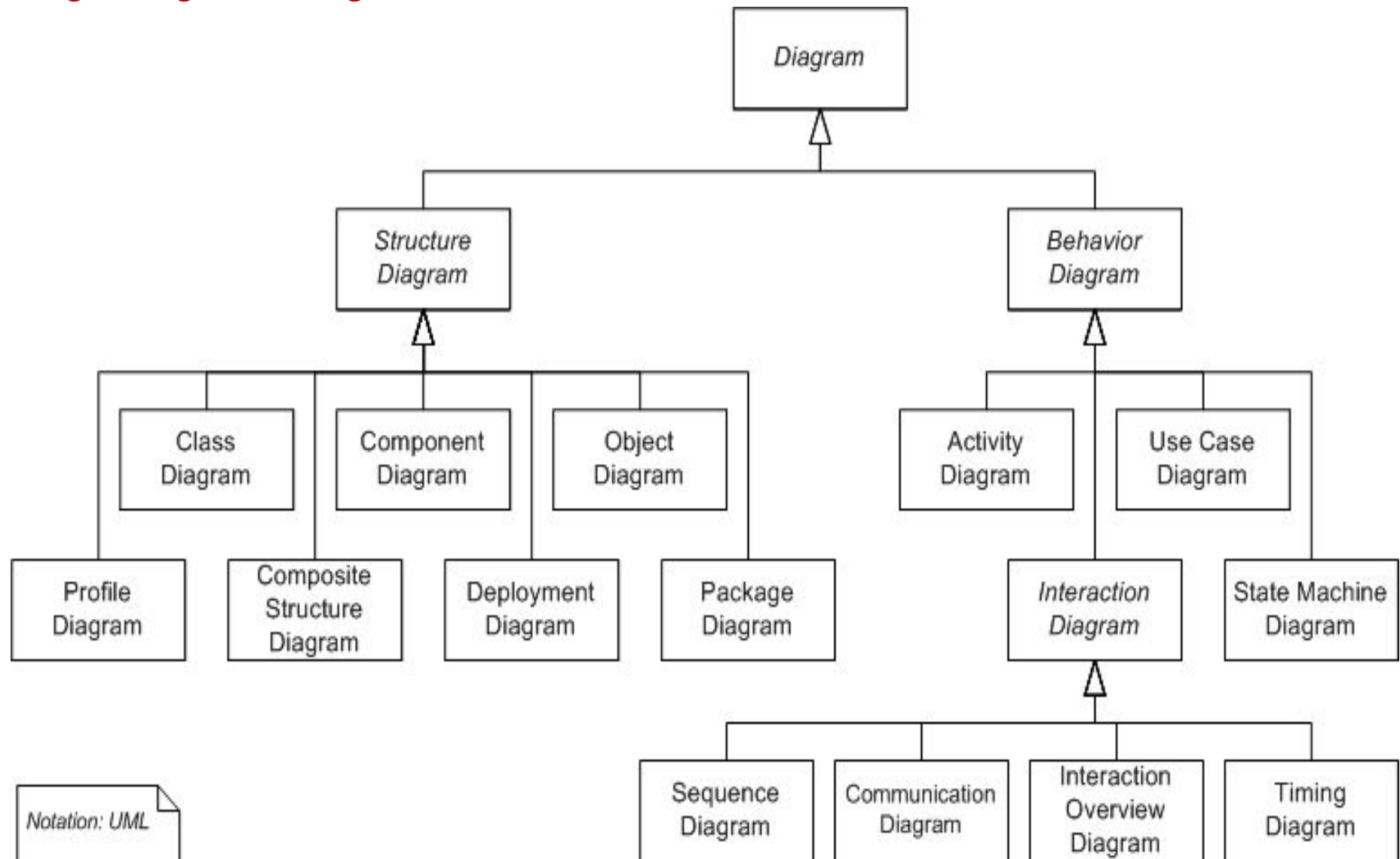
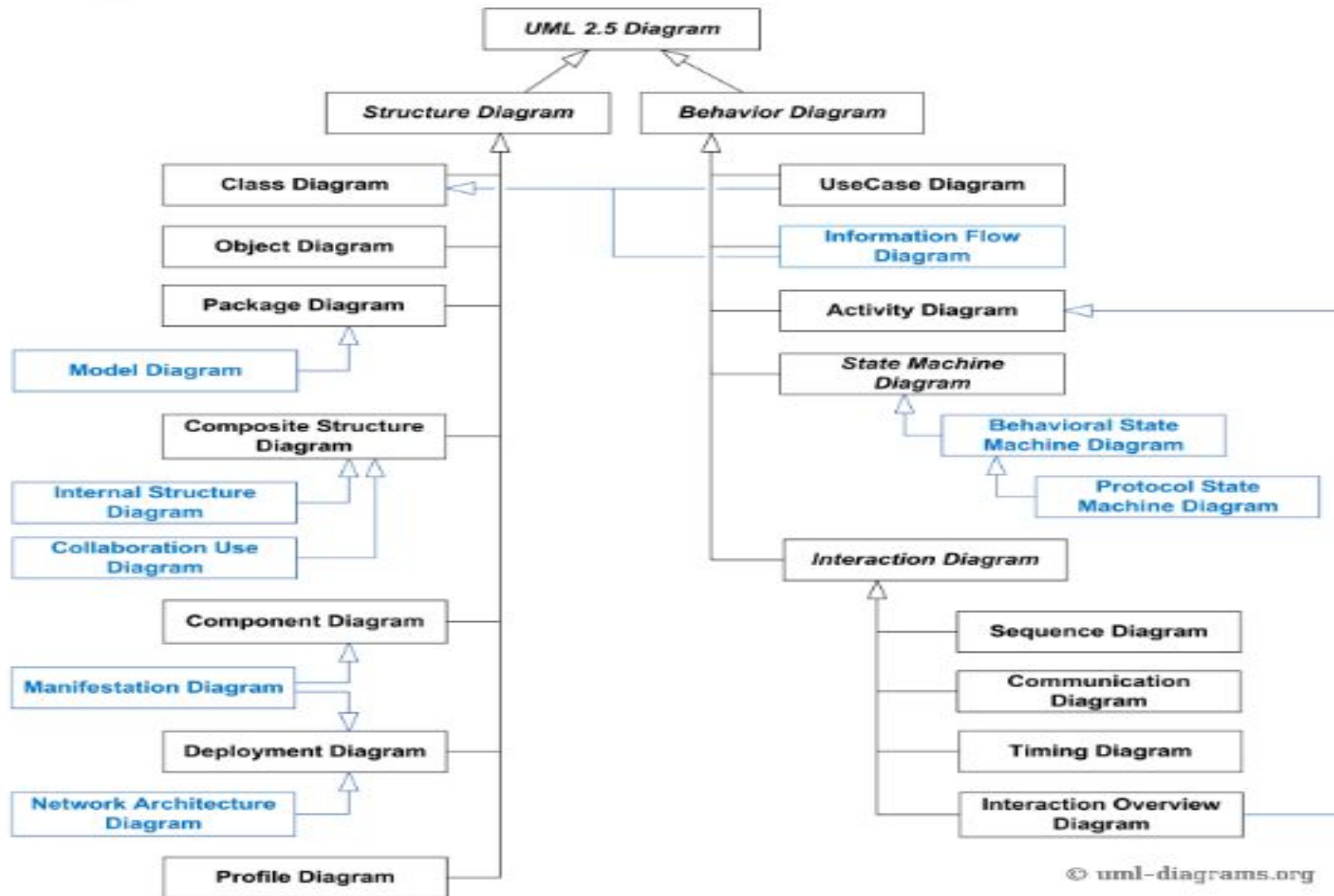


Chart summarizing UML classification

Categorizing UML Diagrams



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UML 2.5 Diagrams Overview.

Note, items in blue are not part of official taxonomy of UML 2.5 diagrams.

*Chart
summarizing
UML
classification for
Version 2.5*

UML Use Case Diagrams, Description And Scenarios

UML – USE CASE DIAGRAMS

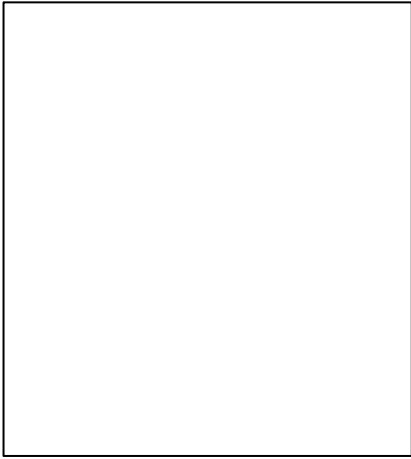
What is a Use Case diagram?

- A use case diagram is used to show the functionality provided by a system in terms of actors, their goals -represented as use cases, and any dependencies between them
- It depicts every functionality that an end user can carry out on the system to be developed. In other words, It represents every case of use / action
- It does not have to show any particular sequence or strict dependence. But it should be orderly arranged

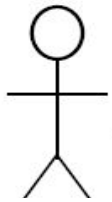
UML – USE CASE DIAGRAMS

Diagram Components

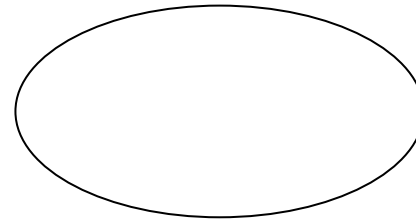
- Boundary



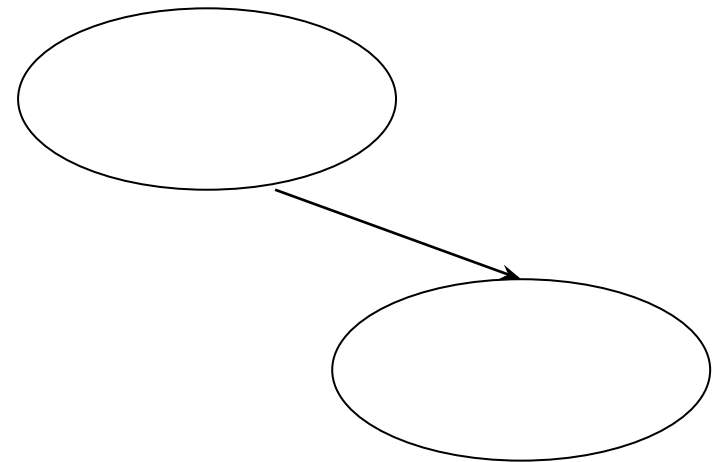
- Actor



- Use Case



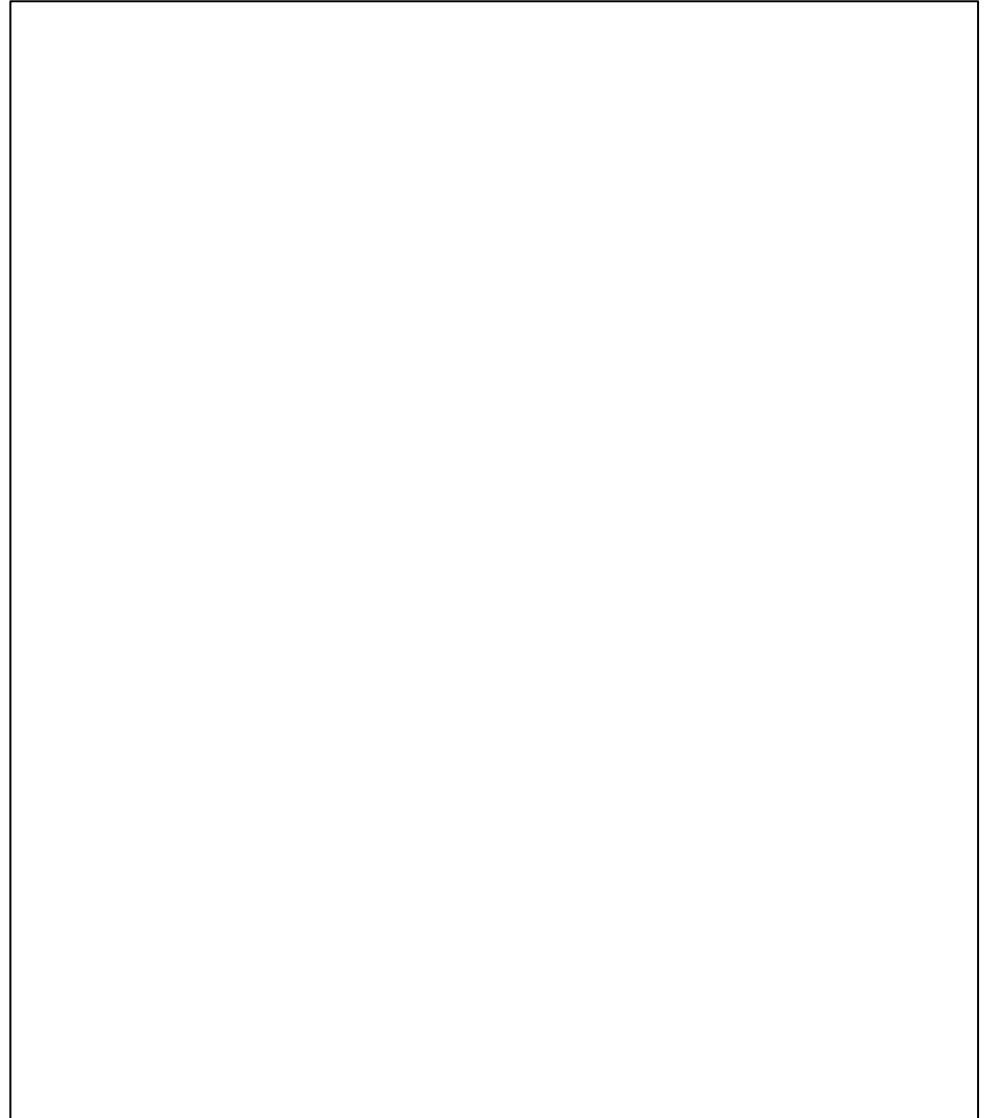
- Relationship



UML – USE CASE DIAGRAMS

Use case boundary

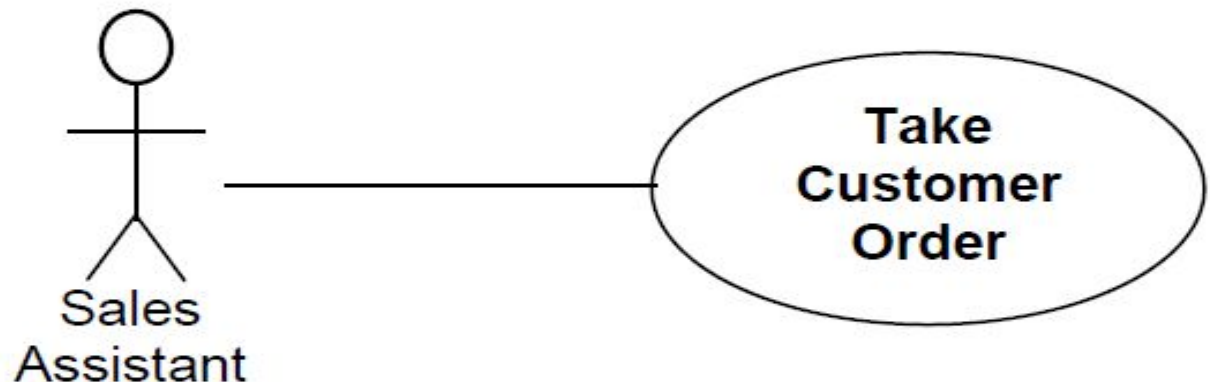
- A rectangular border to contain the behaviour of the system
- It defines the system in question in relation to the external world
- It may represent a sub-system or the system as a whole



UML – USE CASE DIAGRAMS

Who is an Actor?

- In modelling a use case, we use Actors to represent roles that **humans**, hardware devices, or external systems **play** while interacting with a given system, in order to achieve a goal
- For non-human actors to qualify as actors, they must be able to make decisions



UML – USE CASE DIAGRAMS

More on Use Case Actor?

Actors may be both at input and output ends of a use case

Primary Actors use the system to achieve a goal.
To a large extent they initiate the use case

Secondary Actors are those that the system needs assistance from to achieve the primary actor's goal.

UML – USE CASE DIAGRAMS

Use cases and their Scenarios

Use Cases (“Cases of Use”)

- An abstraction of the dialogue between an actor and a system.
- They represent goals that actors achieve on the software system
- It states the possible interactions without giving the details of each scenario

UML – USE CASE DIAGRAMS

Use cases and their Scenarios

Scenarios

- Specify behaviour of use case by description, not modelling
- A sequence of steps that describes a possible interaction between the system and the actor

UML – USE CASE DIAGRAMS

Use Case Relationships

- Association



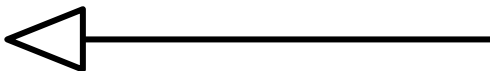
- Include



- Extend



- Generalization



UML – USE CASE (Description & Scenario)

What can a Use case be used for?

A UML Use case diagram may serve as:

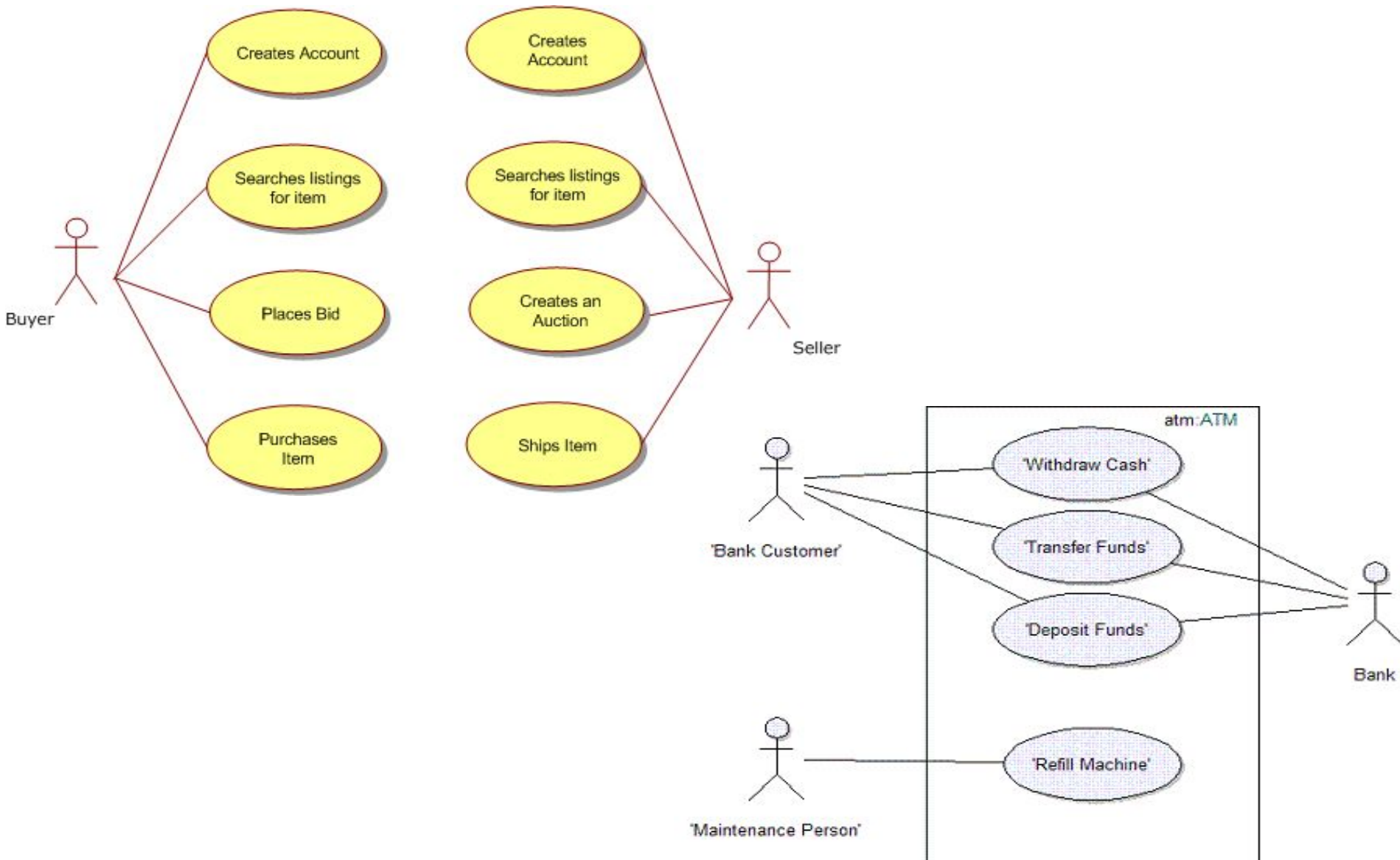
- Means of communication among Software Engineering experts and clients for defining the context of the system
- Definition of software requirements
- Definition of acceptance tests of a system

Further reading:

[http://worldtracker.org/media/library/How-To/For%20Dummies%20eBook%20Collection/UML%202%20for%20Dummies%20\(2003\).pdf](http://worldtracker.org/media/library/How-To/For%20Dummies%20eBook%20Collection/UML%202%20for%20Dummies%20(2003).pdf)

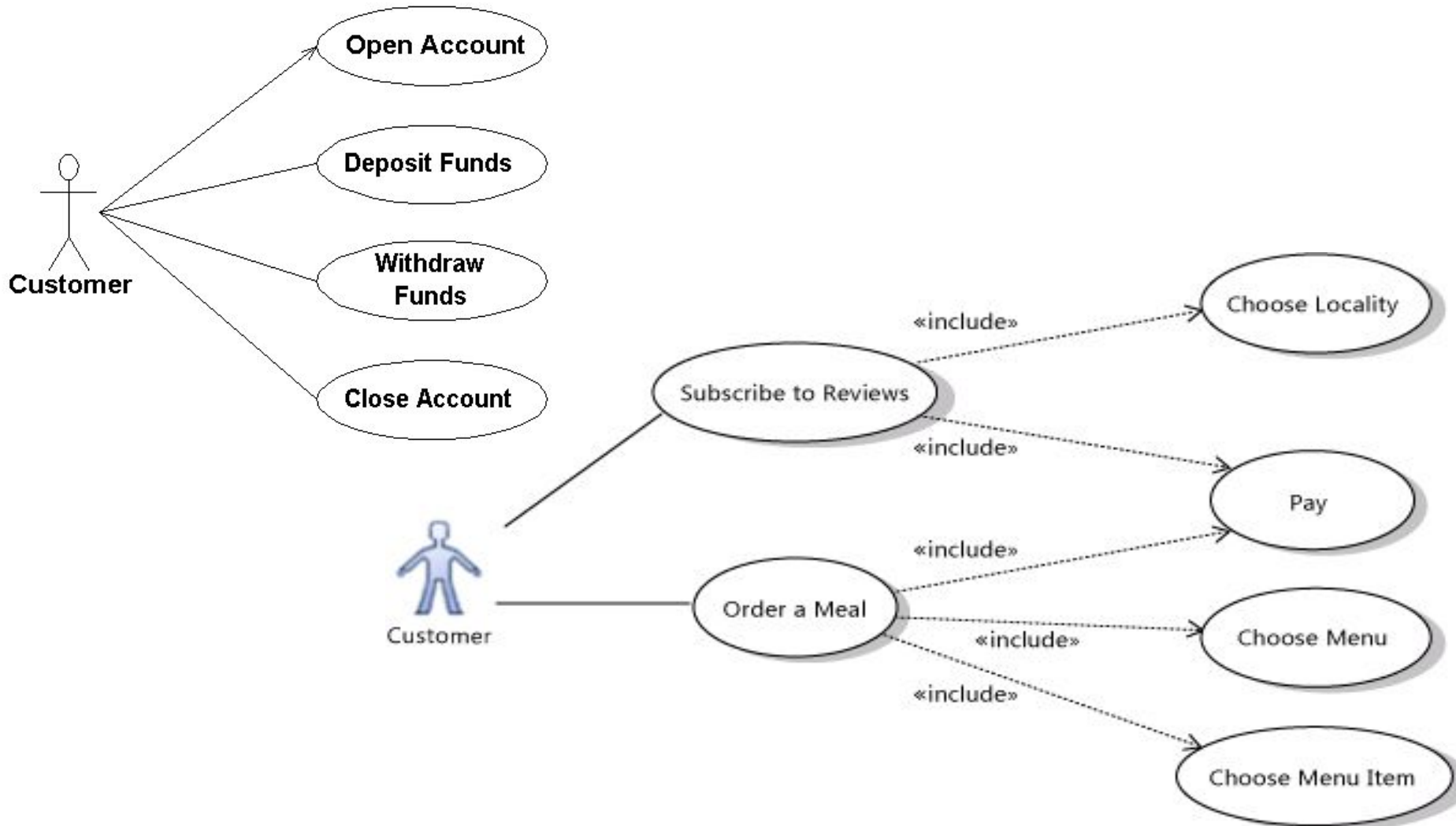
UML – USE CASE DIAGRAMS

Sample use case diagrams



UML – USE CASE DIAGRAMS

Sample use case diagrams



UML – USE CASE DIAGRAMS

Advantages of Use case models

- It is highly user-centric, therefore encouraging the relevant capture of all user concerns
- It can serve as a feeder to other modelling diagrams, and activities during the project
- It contains less text, hence development intentions can be communicated in structured summaries and diagrams
-
- It identifies external and internal factors influencing the system

UML – USE CASE DIAGRAMS

Guidelines:

- Use case modelling styles:
 1. An “actors first” approach: first identify all actors, then link each actor to the use cases
 2. Identifying one actor and the use cases that they’re involved with first and then evolve the model from there.
- Both approaches work. The important point is that different people take different approaches so you need to be flexible.

UML – USE CASE DIAGRAMS

Guidelines:

- Use cases are not necessarily mapped one-to-one to requirements
 - Each requirement must be covered by at least one use case
 - However, use cases may contain many requirements
- Use scenarios to represent assumptions and define system scope
- To name the use cases, start with a verb to show the action that must be performed.
 - E.g. *create account, transfer credit*

UML – USE CASE DIAGRAMS

Guidelines:

- **Start by identifying as many actors as possible.** You should ask how the actors interact with the system to identify an initial set of use cases.
- **On the diagram, you connect the actors with the use cases with which they are involved.**
- Once you notice similarities between use cases, or between actors, start indicating the appropriate relationships between them.
 - i.e. <<includes>> or <<extends>>
 - “Always” or “Sometimes”

The include use case cannot stand alone and the original use case is not complete without the “included one”.

UML – USE CASE DIAGRAMS

Class Exercise:

A student portal for a multinational secondary school in Banana Island, Lagos.

A car park management system

UML – USE CASE (Description & Scenario)

Example 1

Use Case Name	Withdraw Money
Primary Actor	Customer
Secondary Actor(s)	Bank Accounting System
Summary	Customer withdraws cash from the ATM by inserting an ATM card, and entering a valid PIN. Validation is done by the Bank Accounting System
Pre-condition(s)	1. Bank Accounting System is available 2. Money is in the ATM
Normal Flow of Events	1. Customer inserts ATM card 2. ATM reads Bank ID and Account number, and validates with Bank Accounting System 3. Customer enters valid PIN within 10seconds 4. Customer selects account type 5. Customer selects amount to be withdrawn - -

UML – USE CASE (Description & Scenario)

Example 1 Cont'd

Normal Flow of Events Cont'd	- - 6. ATM validates withdrawal amount with Bank Accounting system 7. ATM dispenses cash and ATM card 8. Customer takes cash 9. Customer takes ATM card
Extensions	1. Customer does not enter PIN within 10seconds 2. Customer enters invalid PIN 3. Insufficient Account balance
Post-condition(s)	1. Account balance is updated 2. ATM's Money Inventory is updated

UML – USE CASE (Description & Scenario)

Example 2

Use Case Name	Pay for Job Posting
Primary Actor	Recruiter
Summary	A recruiting company pays for a job posting after uploading the job information, so as to allow job seekers view the vacancy
Pre-condition(s)	Job information uploaded but can not be viewed by Job seekers
Normal Flow of Events	1. Recruiter submits bank card details 2. System validates bank card 3. System charges bank card of the job posting amount 4. Job posting status is made viewable 5. Recruiter is given a payment reference number
Alternate Flow of Events	2a. Bank card type is not accepted by the system 2a.1. The system notifies the user and lists the card types supported 2b. Bank card expired 2b.1. The system notifies the user to use another card 2c. Bank card number is invalid 2c.1. The system notifies the user -

UML – USE CASE (Description & Scenario)

Example 2 Cont'd

Alternate flow Cont'd	- - 3a. Bank card does not have sufficient balance to cover the job posting amount 3a.1. The system notifies the user and requests for another card
Post-condition(s)	1. Job is posted and made viewable by job seekers 2. Recruiter's Bank card is charged

UML – USE CASE (Description & Scenario)

Example 3

Use Case Name	Withdraw cash	
Primary Actor	Customer	
Summary	A customer can withdraw cash from account. He/she may choose from a set of default amounts or enter an amount in thousands	
Pre-condition(s)	No ongoing ATM transaction	
Flow of Events	Customer	ATM
	1. Insert ATM card 4. Enter PIN 7. Choose a default amount	2. Validate card information 3. Request card PIN 5. Validate PIN 6. Display default withdraw amount 8. Dispense cash and eject card
Extensions	5a. Invalid PIN 5a1. System reports error and asks for valid PIN 7a. Customer selects other amount 7a.1 customer enters amount 7a.2 Systems validates that amount is in thousands	
Post-condition(s)	Customer account is updated	

UML – Sequence Diagram

What is a UML Sequence diagram?

- It represents a sequence of steps to achieve a use case goal can be described in several ways.
- A Sequence Diagram focuses on the Message interchange between a number of Lifelines.
 - A Lifeline represents a process time-line, where time increases down the line (with no precise measurement of time).
- Sequence diagram is a variant of the universal Interaction diagrams. Other variants are Communication Diagrams, Interaction Overview Diagrams, and Timing Diagrams
 - Sequence Diagram is the most commonly used variant

UML – Sequence Diagram

Drawing UML Sequence diagram?

- One option is to draw a sequence diagram that shows the interactions between the users and the system's major components
 - Sequence diagrams are used to depict the interactions between system objects or entities in the order of occurrence.
- Another option is to represent textual use case scenario pictorially using a sequence diagrams

UML – Sequence Diagram

Guidelines

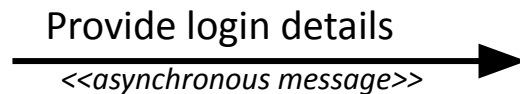
- Parallel vertical lines(lifelines) represent different processes or objects involved
- Horizontal arrows show the messages exchanged between them, in the order in which they occur.
 - This allows the specification of simple runtime scenarios in a graphical manner.
- If the lifeline is that of an object, it demonstrates a role.

UML – Sequence Diagram

Guidelines

To show interaction with messages use horizontal arrows with the message name written above.

Solid arrows for message calls



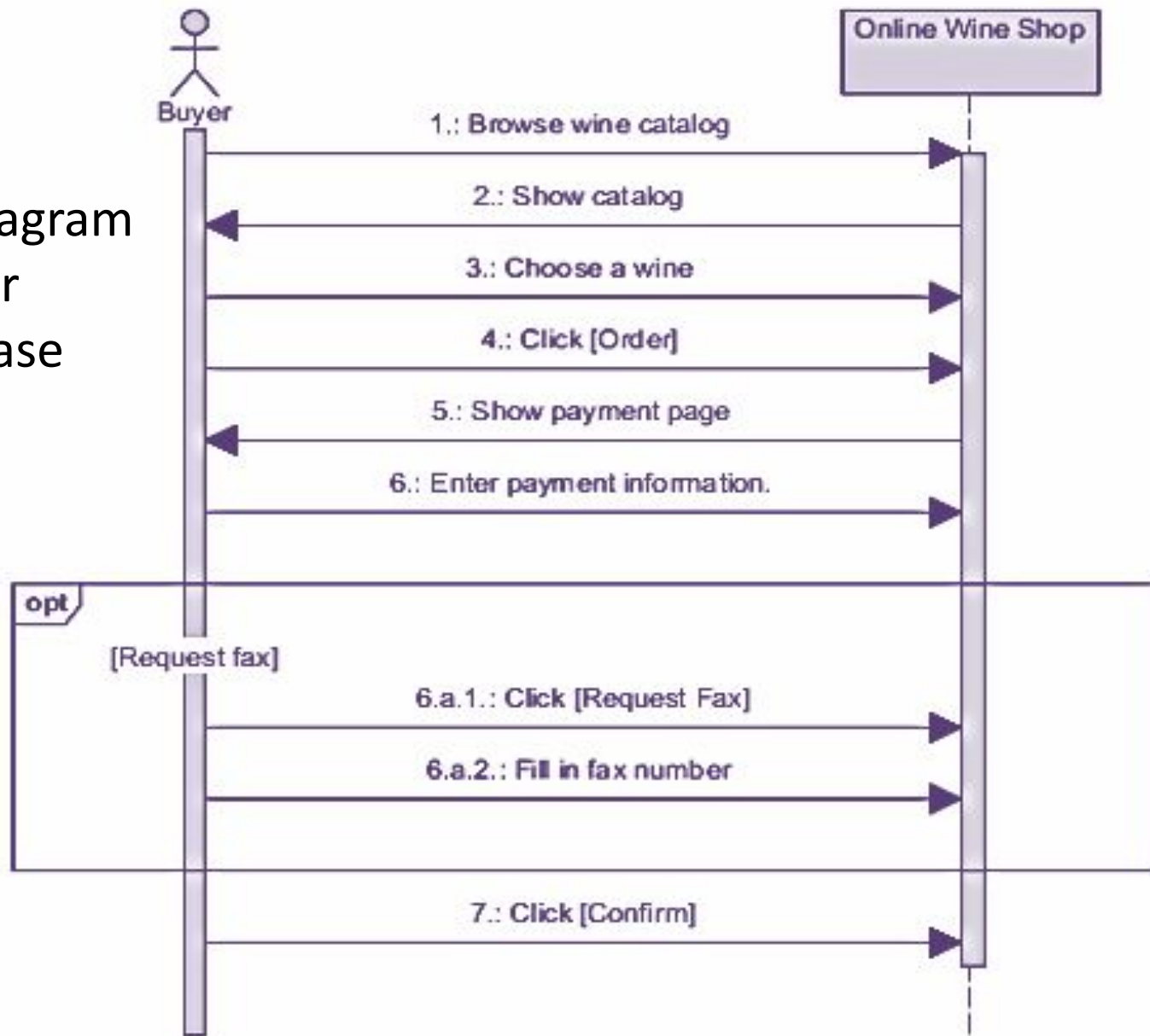
Dashed arrows with stick heads are return messages



UML – Sequence Diagram

Examples

Sequence diagram
for an “Order
Wine” use case



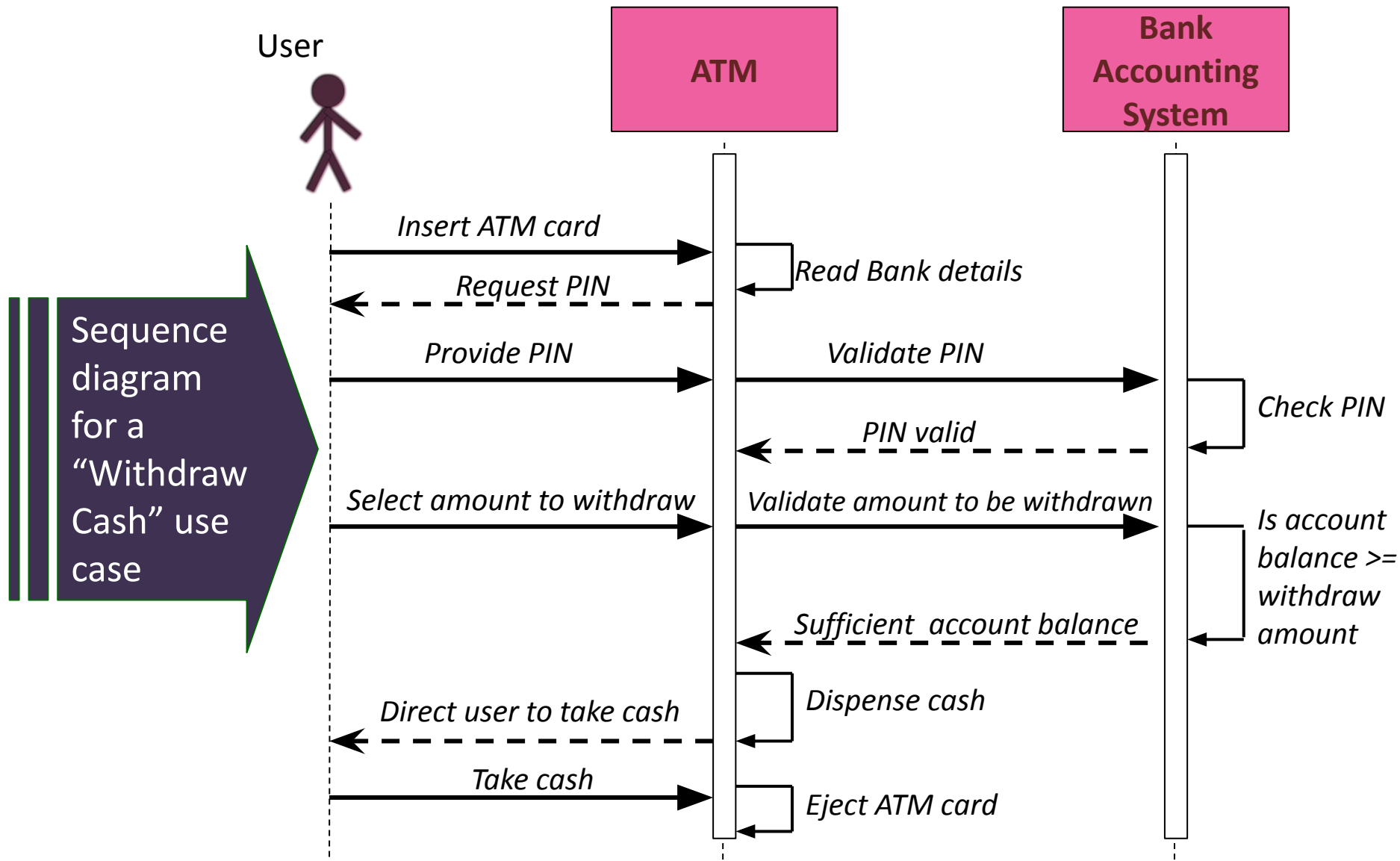
UML – Sequence Diagram

Example from previous use case scenario

Normal Flow of Events	1. Customer inserts ATM card 2. ATM reads Bank ID and Account number, and validates with Bank Accounting System 3. Customer enters valid PIN within 10seconds 4. Customer selects account type 5. Customer selects amount to be withdrawn 6. ATM validates amount to be withdrawn with Bank Accounting system 7. ATM dispenses cash and ATM card 8. Customer takes cash 9. Customer takes ATM card
Extensions	1. Customer does not enter PIN within 10seconds 2. Customer enters invalid PIN 3. Insufficient Account balance
Post-condition(s)	1. Account balance is updated 2. ATM's Money Inventory is updated

UML – Sequence Diagram

Examples



UML – State Diagrams

What is a UML State diagram?

- A State diagram is a type of Behaviour Diagram that displays states and state transitions
- A state diagram describes ALL the possible states that an object can have, the events under which an object changes state (transitions), and the conditions that must be fulfilled before the transition will occur (guards).
- It is also referred to as a State machine diagram, State chart diagram or State transition diagram.

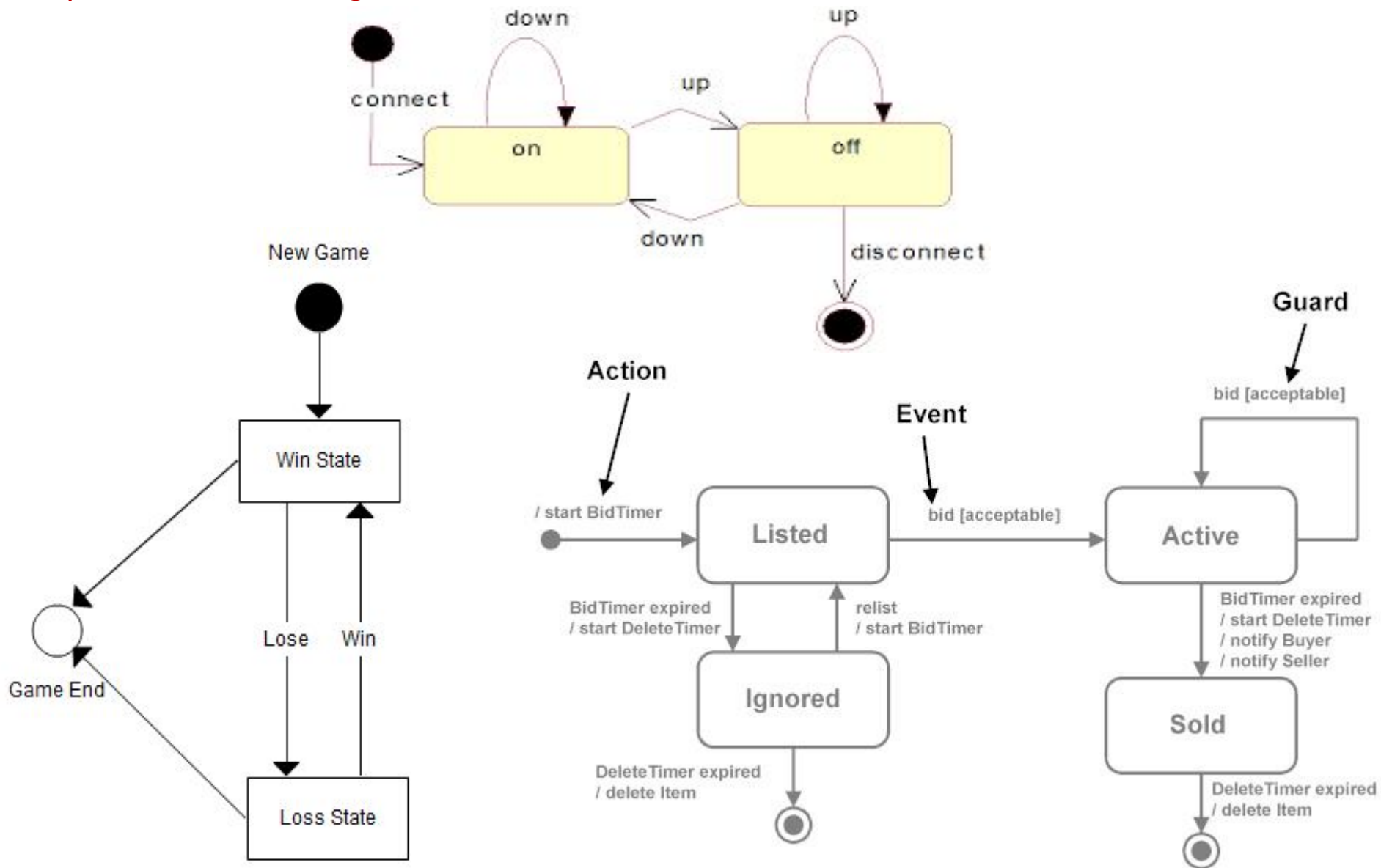
UML – State Diagrams

Why use State diagrams

- State diagrams can be used to describe the behaviour of individual objects over the full set of use cases that affect those objects.
- However, they cannot be used to describe the collaboration between objects that cause the transitions
- State diagrams should only be used for key classes where it is necessary to understand the behaviour of the object through-out the entire system.
 - Not all classes will require a state diagram.
 - E.g. In a University registration system, a state diagram will be required for the Student class

UML – State Diagrams

Sample UML State diagrams



UML – State Diagrams

Definition of terms

- ***What is a State?***

A status during the life of an object. It often satisfies a condition, performs an action, or awaits an event

- ***What is a Transition?***

The change of state within an object.

- ***What is an Event?***

It is an incidence that triggers a state transition

- ***What is a Guard?***

It is a Boolean expression. It makes an event to cause a transition when its value is true

Append “/action-name”
to the “event[guard]”

- ***What is an Action?***

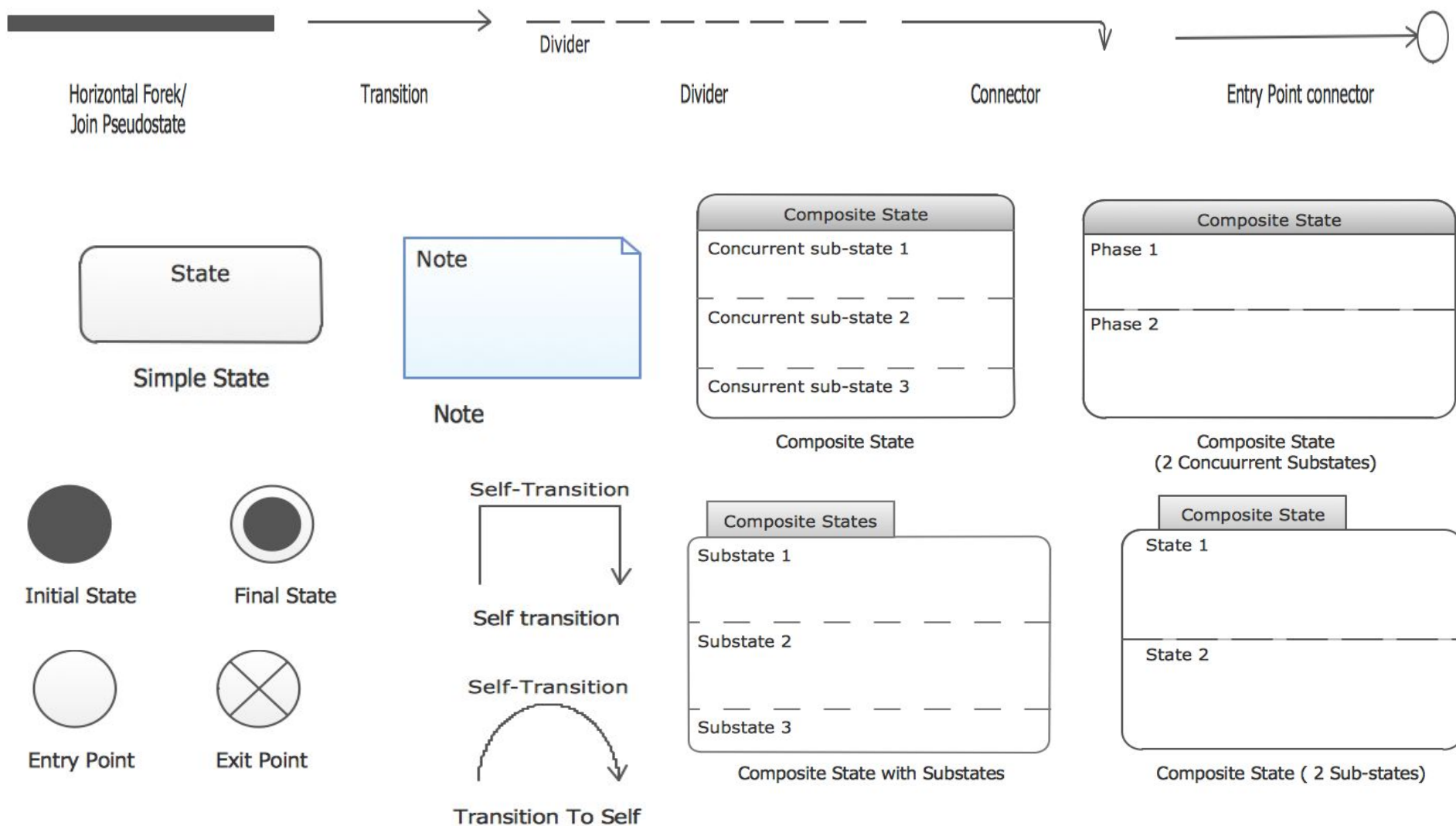
UML – State Diagrams

Guidelines:

- Identify the unique object whose state transition you want to model
- Identify the start state (initial) of the object
- Think about all possible states that the object can be in
- Then start to map out the events that will make the transitions happen
- Remember to label the states, the events and the transitions appropriately

UML – State Diagrams

Guidelines on the required icons



UML – State Diagrams

